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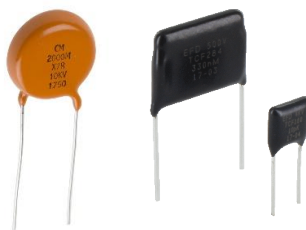
Lab 3 preparation:

Voltage inverter:

1. Which type capacitors are best suited for this circuit? why?

Low-ESR ceramic capacitors (example: X7R or X5R types):

Charge pumps operate at relatively high switching frequencies, and low ESR (Equivalent Series Resistance) capacitors minimize energy losses and heat generation. Ceramic capacitors also have high stability and low parasitic inductance, which are critical for efficient energy transfer in switching circuits like this.



Step down:

1. Calculate R1, R2, L1 and Cout. Check the datasheet for all the information and the appropriate calculation for the adjustable output version

$$V_{out} = V_{ref} \times \left(1 + \frac{R_2}{R_1}\right)$$

From the data sheet page 9

$$V_{ref} = 1.23 \text{ V}$$

For $V_{out} = 9 \text{ V}$

$$R_2 = 6.31 R_1$$

$$\text{If } R_1 = 1 \text{ k}\Omega \rightarrow R_2 = 6.31 \text{ k}\Omega$$

Inductor selection (L1)

$$E \times T = (V_{in} - V_{out}) \times t$$

$$E \times T = (15 - 9) \times \frac{9}{15} \times \frac{1000}{52} [V \times \mu s]$$

$$E \times T = 70 V \times \mu s$$

$$I_{max} = 0.8 \text{ A}$$

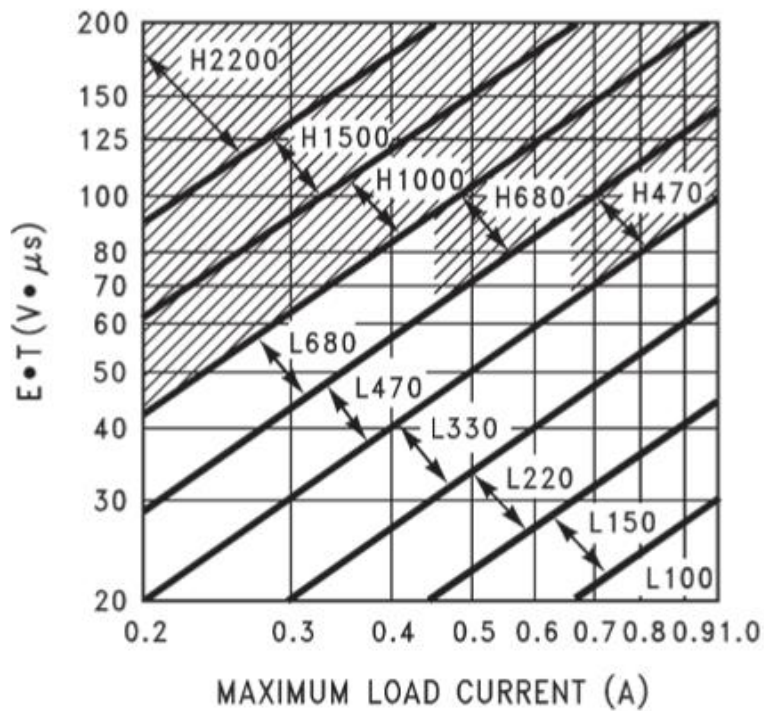


Figure 12. LM2575 Inductor Selection

From the graph we can see **L1= 330 μH**

Calculate capacitor

$$C_{OUT} \geq 7758 \frac{V_{IN(Max)}}{V_{OUT} \cdot L1(\mu H)} (\mu F)$$

$$C_{out} \geq 7758 \times \frac{15}{9 \times 330} \mu F$$

2. How much should the voltage/current ratings be for the inductor, capacitor and diode

The inductor's current rating should be **at least 1.15 times** the maximum output current to account for ripple current and avoid saturation.

$$I_{inductor} = 1.15 \times 0.8 A \approx 0.92 A$$

Voltage Rating: The capacitor should have a voltage rating that is at least **1.5 times** the output voltage to handle any voltage spikes safely. Since the output voltage is 9V, a capacitor with a rating of **16V or 25V** is ideal.

Voltage Rating: The diode's voltage rating should be at least **1.3 times** the input voltage. For a 15V input, you should choose a diode with a rating of at least **20V**.

3. How much current should the lab supply (15V) deliver if the output of the converter is 9V/ 0.8A?

$$P_{in} = \frac{P_{out}}{\eta}$$

η is the efficiency of the converter

$$P_{out} = V_{out} \times I_{out} = 9 V \times 0.8 A = 7.2 W$$

Assuming an efficiency (η) of 80% (0.8):

$$P_{in} = \frac{P_{out}}{\eta} = \frac{7.2}{0.8} = 9 W$$

$$I_{in} = \frac{P_{in}}{V_{in}} = \frac{9 W}{15 V} = 0.6 A$$

4. What is continuous and discontinuous mode in a DC – DC converter?

Can you explain one of the consequences of discontinuous mode for a step-down converter looking at this graph (d = duty cycle)

In discontinuous mode, the converter's performance degrades, especially at low currents. The output voltage drops more than expected, and the overall efficiency decreases. Therefore, converters typically aim to operate in continuous mode at higher loads for better voltage regulation and efficiency.